

# Samuel Leblanc

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## Education

### Queen's University

2025 - 2027

*M.Sc. in Mathematics*

Research project: Cluster algebras, generalized cluster categories, and representation theory

Supervisors: [Ivan Dimitrov](#), [Charles Paquette](#), and [David Wehlau](#)

### Université de Sherbrooke

2022 - 2025

*B.Sc. in Mathematics*

Mention of Excellence – Faculty Dean's List, Winter 2025

## Research Interests

Representation theory of (finite-dimensional and associative) algebras, cluster algebras, homological algebra.

## Papers

- Brüstle, T., Desrochers, J., Leblanc S., *Generalized Rank via Minimal Subposet*. [arXiv:2510.10837](#)  (2025)
- Leblanc, S., Rasolomanana, A., Armenta, M., *Hidden Activations Are Not Enough: A General Approach to Neural Network Predictions*. [arXiv:2409.13163](#)  (2024)
- Armenta, M., Leblanc, S., *Batalin-Vilkovisky structure on Hochschild cohomology with coefficients in the dual algebra*. [arXiv:1810.13023](#)  (2018) (I joined as coauthor in 2025; revisions and extensions)

## Student Papers

- Baril, L., Leblanc, S., *Multiplicité de la représentation indicatrice*. Submitted to: [Cahiers mathématiques de l'Université de Sherbrooke](#)  (2025)   
Supervisor: [Thomas Brüstle](#)
- Leblanc, S., *Dégénération des représentations de carquois de type  $A_3$  à deux puits*. To appear in: [Cahiers mathématiques de l'Université de Sherbrooke](#)  (2024)   
Supervisors: [Emily Cliff](#) and [Shiping Liu](#)
- Leblanc, S., *Transformations de cercles orientés tangents sur la sphère de Riemann*. Submitted to: [Cahiers mathématiques de l'Université de Sherbrooke](#)  (2023)   
Supervisor: [Jean-Philippe Burelle](#)

## Posters

- Minimal Initial Subcategory*. Presented at Queen's Mathematics Summer School and Modern Methods in Representation Theory (2026)   
★ This poster won a *Best Poster Award* at Queen's Mathematics Summer School.
- Multiplicity of the Interval Module*. Presented at AARMS-CMS Student Poster Session (CMS Summer Meeting) (2025)   
Collaborators: Laurianne Baril and Justin Desrochers. Supervisor: [Thomas Brüstle](#).
- Kernel of the Rank Invariant* (with Justin Desrochers). Presented at Summer Research School, Applications of Representation Theory in Topological Data Analysis and Geometric Invariant Theory (2024) 

## Software

### knowledgematrix

 GitHub

- A Python library for implementing neural networks and computing their associated *knowledge matrices* (i.e.,  $N_V$  (Lemma 7.4) in [this paper](#)  and  $M(W, f)(x)$  in [this paper](#) .

- Tools: Python

#### **simple\_adversarial\_detection**

 GitHub

- Very simple version of the code used for the experiments in the paper Hidden Activations Are Not Enough: A General Approach to Neural Networks Predictions. [arXiv:2409.13163](https://arxiv.org/abs/2409.13163) 
- Tools: Python

#### **upperhalfplane**

 GitHub

- Visualize the action of  $\mathrm{PSL}(2, \mathbb{R})$  on the upper half plane (Poincaré half plane model) interactively. [samueleblanc.com/software/upperhalfplane](https://samueleblanc.com/software/upperhalfplane) 
- Tools: CindyJS, JavaScript, HTML, CSS

#### **riemannsphere**

 GitHub

- Visualize the action of  $\mathrm{PSL}(2, \mathbb{C})$ , i.e., Möbius transformations, and  $\mathrm{PSP}(4, \mathbb{R})$  on the Riemann sphere interactively. [samueleblanc.com/software/riemannsphere](https://samueleblanc.com/software/riemannsphere) 
- Supervisor: [Jean-Philippe Burelle](#).
- Tools: CindyJS, JavaScript, HTML, CSS

#### **MatTalX**

 GitHub

- Chrome Extension and Firefox Add-on that allow the user to convert LaTeX commands into plain text, enabling them to write symbols anywhere. <https://mattalx.org> 
- Tools: JavaScript, HTML, CSS, Bash

## **Talks**

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1. *(Co)limit Computation and Its Application to Representation Theory* (January 26, 2026)  
Algebra & Geometry Seminar, Queen's University
2. *Generalized Rank in Generalized Linear Algebra* (November 12, 2025)  
Grad Seminar, Department of Mathematics and Statistics, Queen's University
3. *Analyse topologique de données* (February 13, 2025)  
Club mathématiques de l'Université de Sherbrooke
4. *La propagation avant en tant que matrice* (November 14, 2024)   
Club mathématiques de l'Université de Sherbrooke
5. *Visualisation de transformations sur la sphère de Riemann* (March 21, 2024)   
Club mathématiques de l'Université de Sherbrooke
6. *Théorie des représentations des réseaux de neurones* (October 5, 2023)  
Club mathématiques de l'Université de Sherbrooke

## **Teaching**

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### **Guest Lecturer**

#### **Real Analysis (MATH/MTHE 328)**

**Winter 2026**

Replaced the course instructor for one week. Delivered 3 hours of lectures.  
*Department of Mathematics and Statistics, Queen's University*

### **Teaching Assistant**

#### **Differential and Integral Calculus (MATH 128)**

**Winter 2026**

*Department of Mathematics and Statistics, Queen's University*

#### **Méthodes quantitatives en communication marketing (MQG301)**

**Winter 2025**

*École de gestion, Université de Sherbrooke*

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| <b>Calcul vectoriel (MAT298)</b><br><i>Département de mathématiques, Université de Sherbrooke</i>                                                                                                                                                                                                                   | <b>Fall 2024</b>   |
| <b>Statistique appliquée à la gestion (MQG222)</b><br><i>École de gestion, Université de Sherbrooke</i>                                                                                                                                                                                                             | <b>Summer 2024</b> |
| <b>Statistique appliquée à la gestion (MQG222)</b><br><i>École de gestion, Université de Sherbrooke</i>                                                                                                                                                                                                             | <b>Winter 2024</b> |
| <b>Grader</b>                                                                                                                                                                                                                                                                                                       |                    |
| <b>Differential and Integral Calculus (MATH 121B/128)</b><br><i>Department of Mathematics and Statistics, Queen's University</i>                                                                                                                                                                                    | <b>Winter 2026</b> |
| <b>Real Analysis (MATH/MTHE 328)</b><br><i>Department of Mathematics and Statistics, Queen's University</i>                                                                                                                                                                                                         | <b>Winter 2026</b> |
| <b>Algebraic Structures (MTHE 217)</b><br><i>Smith Engineering, Queen's University</i>                                                                                                                                                                                                                              | <b>Fall 2025</b>   |
| <b>Calculus I (APSC 171)</b><br><i>Smith Engineering, Queen's University</i>                                                                                                                                                                                                                                        | <b>Fall 2025</b>   |
| <b>Differential and Integral Calculus (MATH 121)</b><br><i>Department of Mathematics and Statistics, Queen's University</i>                                                                                                                                                                                         | <b>Fall 2025</b>   |
| <b>Mathématiques discrètes (MAT120)</b><br><i>Département de mathématiques, Université de Sherbrooke</i>                                                                                                                                                                                                            | <b>Fall 2024</b>   |
| <b>Tutor</b>                                                                                                                                                                                                                                                                                                        |                    |
| <b>Mathematics, 10th grade</b><br><i>Volunteering with <a href="#">Le Diplôme avant la Médaille</a> </i>                                                                                                                         | <b>2023 - 2024</b> |
| <b>Algèbre linéaire et géométrie vectorielle (MAT902)</b><br><i>Université de Sherbrooke</i>                                                                                                                                                                                                                        | <b>Summer 2023</b> |
| <b>Biomécanique humaine (KIN325)</b><br><i>Université de Sherbrooke</i>                                                                                                                                                                                                                                             | <b>Winter 2023</b> |
| <b>Service</b>                                                                                                                                                                                                                                                                                                      |                    |
| <b>Academic Coordinator, Queen's GMS</b><br>During the 2026-2027 academic year, Trevor Shillington and I will be the academic coordinators of the Graduate Mathematics and Statistics Society at Queen's University. We will organize the Graduate Student Seminar of the Department of Mathematics and Statistics. | <b>2026 - 2027</b> |
| <b>Selected Academic Activities</b>                                                                                                                                                                                                                                                                                 |                    |
| <b>Modern Methods in Representation Theory</b><br>Attended the Modern Methods in Representation Theory Conference at the University of Ottawa (Ottawa, ON). July 7 to 10, 2026.                                                                                                                                     | <b>Summer 2026</b> |
| <b>Queen's Mathematics Summer School</b><br>Attended the Queen's Mathematics Summer School at Queen's University (Kingston, ON). June 22 to 26, 2026.                                                                                                                                                               | <b>Summer 2026</b> |
| <b>Auslander Conference</b><br>Attended the Maurice Auslander Centenary Lectures and International Conference at the Oceanographic Institution (Woods Hole, MA, USA). April 22 to 27, 2026.                                                                                                                         | <b>Winter 2026</b> |
| <b>35th RTA Meeting</b><br>Attended the 35th Meeting on the Representation Theory of Algebras and Related Topics at the Université de Sherbrooke (Sherbrooke, QC). October 23 to 25, 2025.                                                                                                                          | <b>Fall 2025</b>   |

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| <b>Route 81 Conference</b><br>Attended the Route 81 Conference at Queen's University (Kingston, ON). September 27, 2025.                                                                                                        | <b>Fall 2025</b>   |
| <b>Canadian Mathematical Society (CMS) Meeting</b><br>Attended the 2025 CMS Summer Meeting at the Université Laval (Québec, QC). June 7 to 9, 2025.                                                                             | <b>Summer 2025</b> |
| <b>34th RTA Meeting</b><br>Attended the 34th Meeting on the Representation Theory of Algebras and Related Topics at the Université de Sherbrooke (Sherbrooke, QC). October 4 and 5, 2024.                                       | <b>Fall 2024</b>   |
| <b>Research School</b><br>Attended the Summer Research School: Applications of Representation Theory in Topological Data Analysis and Geometric Invariant Theory, at the UQAM (Montréal, QC). June 3 to 7, 2024.                | <b>Summer 2024</b> |
| <b>Introduction to Research (MAT523): Topological Data Analysis</b><br>Optional course. <i>Département de mathématiques, Université de Sherbrooke</i><br>Supervisor: <a href="#">Thomas Brüstle</a>                             | <b>Winter 2024</b> |
| <b>BIRS Workshop</b><br>Assisted (online) to the BIRS Workshop: Representation Theory and Topological Data Analysis. April 8 to 11, 2024.                                                                                       | <b>Winter 2024</b> |
| <b>Research internship: Representation Theory of Quivers</b><br><i>Département de mathématiques, Université de Sherbrooke</i><br>Supervisors: <a href="#">Emily Cliff</a> and <a href="#">Shiping Liu</a>                       | <b>Summer 2023</b> |
| <b>Experimental Mathematics Lab (MAT001): Projective Geometry</b><br>Course taken beyond B.Sc. requirements. <i>Département de mathématiques, Université de Sherbrooke</i><br>Supervisor: <a href="#">Jean-Philippe Burelle</a> | <b>Winter 2023</b> |

## Languages

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French (native), English (advanced).